Supplementary material

Separating decision-rule failures from representation failures in external validation of diabetic retinopathy grading models

Ahmadreza Nourozi, Alireza Chegeni

Contents

Supplementary Tables S1–S8 **2**

Supplementary Note 1. Decision rules **6**

Supplementary Note 2. Macular oedema: a supervision ceiling and an underivable grade **6**

Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure **7**

Supplementary Note 4. Correction of an archived ordinal-head comparison **7**

Supplementary Note 5. Seed variation (O1–O3) **7**

Supplementary Note 6. Reverse direction (O4–O6) **8**

Supplementary Note 7. Content-overlap audit **9**

Full decision-rule matrices **10**

Supplementary Tables S1–S8

Table S1: Analysis plans and the runs they cover. Commit identifiers and times (UTC) are from the public repository history; launch times are from the Kaggle kernel listing archived in docs/generated/paper/kaggle_runs.csv. Every run listed was launched after the commit of its plan and pinned the code commit it ran. Plans: docs/PAPER_EXPERIMENTS_preregistration.md and docs/PAPER_REVISION_preregistration.md.

Plan	Commit	Committed (UTC)	First run launched (UTC)
Repository history begins (evaluation protocol present)	`5846b1b`	2026-08-31 17:51	–
Part A: derivation test of the macular-oedema grade	`f79abc4`	2026-09-07 17:37	–
CORAL, CORN and ConvNeXt interventions (hypotheses in EXPERIMENTS.md)	`b723945`	2026-09-05 17:10	2026-09-05 17:10
Journal outcomes O1–O9 (journal experiments plan)	`46f1bdd`	2026-10-16 11:03	2026-10-16 11:03
Revision outcomes A1–E2 (revision plan)	`1456815`	2026-10-20 11:21	2026-10-20 11:21

Table S2: Reporting checklist (TRIPOD+AI and CLAIM). Each reporting domain mapped to the place in the article where it is addressed; the journal's checklist forms are completed from this mapping at submission.

Domain	Where and how it is reported
Title and abstract	Title names the task and the finding; the abstract states data, models, external sets, decision rules and main results.
Background and objectives	Introduction, including what this study adds to ORDER-DR and to recent calibration studies.
Data sources	Section 2.2 and Table 1: IDRiD, Messidor-2 (adjudicated grades), EyePACS, APTOS 2019, DDR, DeepDRiD, SUSTech-SYSU.
Participants and eligibility	Every gradable image with a DR grade 0–4; DDR grade 5 (ungradable) excluded. No patient-level demographics are released by the corpora (limitation).
Data preparation	Section 2.3 and Table S3: retinal crop, cache at 560 pixels, resizing, augmentation.
Outcome (reference standard)	International 5-grade DR scale; referable DR is grade $\geq$2, defined by DR alone because macular oedema is not graded externally.
Sample size	Fixed by the public corpora; bootstrap intervals throughout; the local recalibration sample size is an explicit result (Section 3.8).
Missing data	Messidor-2 has no three-grade DME label; the DME loss is masked for those images.
Data partitions	Frozen 5-fold split of the development pool; external sets never used for training, selection or threshold fitting except in the diagnostic R4/R4m; content audit (Note 7).
Model specification	Section 2.3, Table 2 and Table S3.
Model output and decision rule	Section 2.4 and Note 1 (R1–R5b).
Performance measures	QWK, Cohen's κ, accuracy, per-grade recall, precision and F1, referable sensitivity and specificity, per-threshold AUC, Harrell's C, Brier score, ECE, PPV and NPV.
Uncertainty	Paired bootstrap (2 000 resamples) over patients where identified, otherwise images.
Class imbalance	Grade-balanced sampler; per-grade results for every grade.
Fairness	No subgroup attributes are released; between-population behaviour is studied across four external populations and two directions.
Analysis plans	Committed to version control before each analysis (Table S1); falsified outcomes are reported.
Code and data availability	Data and code availability statement.

Domain	Where and how it is reported
Ethics	Ethics statement (public, de-identified data).
Use of AI tools	Section 2.8 (AI-assisted coding) and the generative-AI declaration (editing of the text).
Limitations	Discussion.

Table S3: Configurations. Each configuration changes one factor relative to its control. Dev QWK: development out-of-fold DR QWK at matched calibration (fold-wise cross-fitted cut-points); folds: development folds trained.

Model	Control	Backbone	DR head	Pixels	Pretraining	Epochs	Factor	Dev QWK
E08	–	DenseNet121	cumulative	448	ImageNet + EyePACS (5 ep)	40	reference	0.8646
E09	E08	DenseNet121	cumulative	224	ImageNet + EyePACS	40	resolution	0.8389
E10	E08	DenseNet121	cumulative	640 (560 effective)	ImageNet + EyePACS	40	resolution	0.8749
E17NAT	E10	DenseNet121	cumulative	640	ImageNet + EyePACS	40	native-resolution source files	0.8662
E11FULL	E10	EfficientNet-B3	cumulative	448	ImageNet + EyePACS	30	backbone (selected model)	0.8954
E14MAC	E08	DenseNet121	cumulative	448 + macula crop	ImageNet + EyePACS	40	fovea-centred crop for DME head	0.8595
E15LPFT	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40 (LP 5)	linear probe then fine-tune	0.8655
E20CORAL	E08	DenseNet121	CORAL	448	ImageNet + EyePACS	40	ordinal head	0.8569
E22CORN	E08	DenseNet121	CORN	448	ImageNet + EyePACS	40	ordinal head	0.8564
E21CNXA	E08	ConvNeXt-tiny	cumulative	448	ImageNet + EyePACS	40	backbone (2 folds)	0.8267 (2 f)
I24FULL	E09	RETFound ViT-L/16	cumulative	224	RETFound (colour fundus)	30 (LP 5)	foundation encoder	0.7976
S1E08–S4E08	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	seed only	–
R1APTOS	E08	DenseNet121	cumulative	448	ImageNet + EyePACS	40	trained on APTOS (reverse)	–

Table S4: Pre-registered interventions, shipped and matched calibration. Difference (intervention minus control) in development DR QWK with 95% paired-bootstrap intervals (DME QWK for the macula crop; APTOS DR QWK for the ensemble).

Intervention	Control	Folds	Matched difference [95% CI]	Verdict
EfficientNet-B3 backbone	E10	5	+0.0207 [+0.0105, +0.0320]	confirmed
Macula-centred crop (DME QWK)	E08	5	−0.0237 [−0.0566, +0.0094]	falsified
Linear probe then fine-tune	E08	5	+0.0008 [−0.0105, +0.0126]	falsified
Native 640-px resolution	E10	5	−0.0086 [−0.0199, +0.0025]	falsified
CORAL ordinal head	E08	5	−0.0076 [−0.0184, +0.0032]	falsified
CORN ordinal head	E08	5	−0.0082 [−0.0207, +0.0036]	falsified
ConvNeXt-tiny backbone	E08	2	−0.0362 [−0.0593, −0.0153]	falsified, significantly worse
RETFound ViT-L/16 (fine-tuned)	E09	5	−0.0414 [−0.0583, −0.0238]	falsified, significantly worse
Six-model ensemble (APTOS, external)	E10	5	+0.0025 [−0.0016, +0.0066]	falsified

Table S5: Decision-rule reanalysis of a published model family (ORDER-DR, APTOS→Messidor-2). Mean ± s.d. over the authors' three seeds, computed from their released image-level probabilities on 1,744 Messidor-2 images (270 mild). For ORDER-DR the R2 row uses the authors' own fusion and cut-points and reproduces their reported result. R4/R4m use external labels (two-fold cross-fitting) and are diagnostic bounds, not performance claims.

Model	Rule	QWK	Mild rec. (%)	Mild F1	Sens. (%)	Spec. (%)
CB-384	R1 argmax	0.500 ± 0.018	2.0	0.027	32.8	99.5
	R2 val.-fitted	0.544 ± 0.029	2.5	0.035	38.2	99.0
	R4 ext. QWK	0.672 ± 0.030	27.8	0.231	55.1	95.4
	R4m ext. recall	0.614 ± 0.060	38.4	0.255	72.6	85.2
LORS-384	R1 argmax	0.531 ± 0.014	3.5	0.049	36.3	98.9
	R2 val.-fitted	0.570 ± 0.023	2.7	0.041	42.8	98.7
	R4 ext. QWK	0.683 ± 0.045	30.5	0.245	55.9	96.1
	R4m ext. recall	0.677 ± 0.046	36.5	0.285	67.6	91.3
Swin-T	R1 argmax	0.552 ± 0.045	5.2	0.062	35.4	96.9
	R2 val.-fitted	0.564 ± 0.026	3.3	0.045	43.2	96.2
	R4 ext. QWK	0.674 ± 0.022	13.0	0.147	64.8	93.8
	R4m ext. recall	0.629 ± 0.041	30.2	0.233	75.0	88.9
ORDER-DR	R1 argmax	0.509 ± 0.017	2.7	0.040	34.1	99.2
	R2 val.-fitted	0.642 ± 0.036	50.5	0.277	39.8	99.1
	R4 ext. QWK	0.676 ± 0.025	24.2	0.212	56.7	95.4
	R4m ext. recall	0.638 ± 0.049	42.0	0.284	70.0	88.8

Table S6: Models trained on about 28 000 EyePACS images (outcome E2). Referral area under the curve, mean predicted-minus-true grade and referable sensitivity at the shipped rule (R1), with cut-points refitted on external labels (R4), and with a threshold fitted on the held-out EyePACS fold for 85% sensitivity.

Model (EyePACS-trained)	Corpus	AUC	Pred.−true	Sens. R1	Spec. R1	Sens. R4	Sens. thr. 85
EfficientNet-B3	APTOS	0.938	+0.14	98.9	83.7	96.4	99.9
	DDR	0.952	−0.23	73.8	97.7	80.1	90.8
	Messidor-2	0.956	−0.08	85.3	94.3	77.2	95.8
	IDRiD	0.984	+0.06	96.6	87.0	94.4	99.4
	DeepDRiD	0.969	−0.01	92.9	89.7	91.2	98.7
	SUSTech	0.999	+0.05	98.8	98.5	99.0	100.0

Table S7: Image degradation and image statistics (outcomes A2, A3). Top: change relative to the undegraded re-scoring of the development pool (out of fold) in mean predicted-minus-true grade and in referable sensitivity at the shipped rule. Bottom: median image statistics per corpus (interquartile range in the repository).

Degradation	Reference model		Selected model	
	Δ pred.−true	Δ sens. (pts)	Δ pred.−true	Δ sens. (pts)
JPEG quality 30 at 512 px	−0.45	−36.8	−0.53	−45.6
Downsampled to 256 px	−0.15	−6.2	−0.04	−1.3
Gaussian blur, sigma 2 px	−0.30	−15.4	−0.24	−15.9
Gamma 1.8 (darker)	−0.16	−8.7	−0.09	−6.7
Contrast reduced to 60%	−0.13	−5.1	−0.06	−2.3

Corpus	Long side (px)	JPEG quality	Brightness	Laplacian var.
IDRiD	4,288	50	97	52
Messidor-2	512	95	87	39
APTOS	2,048	–	91	52
DDR	512	95	93	70
DeepDRiD	1,824	95	93	47
SUSTech	2,880	75	120	51

Table S8: What the operating point means for a screening programme. Referable patients missed and false referrals per 10 000 people screened, and predictive values, for the selected model at its shipped rule (R1) and with cut-points refitted on local labels (R4), at three referable prevalences. Sensitivity and specificity are those measured on each external set.

Corpus	Rule	Sens. (%)	Spec. (%)	Prevalence (%)	Missed /10 000	False referrals /10 000	PPV (%)	NPV (%)
APTOS	R1	99.3	84.3	5	4	1,489	25.0	100.0
		99.3	84.3	10	7	1,411	41.3	99.9
		99.3	84.3	20	15	1,254	61.3	99.8
	R4	96.5	86.6	5	17	1,275	27.4	99.8
		96.5	86.6	10	35	1,208	44.4	99.6
		96.5	86.6	20	70	1,074	64.2	99.0
DDR	R1	70.3	98.7	5	149	124	73.9	98.4
		70.3	98.7	10	297	117	85.7	96.8
		70.3	98.7	20	594	104	93.1	93.0
	R4	81.2	94.2	5	94	550	42.5	99.0
		81.2	94.2	10	188	521	60.9	97.8
		81.2	94.2	20	375	463	77.8	95.3
DeepDRiD	R1	90.3	92.3	5	49	728	38.3	99.4
		90.3	92.3	10	97	689	56.7	98.8
		90.3	92.3	20	194	613	74.7	97.4
	R4	91.4	91.1	5	43	845	35.1	99.5
		91.4	91.1	10	86	800	53.3	99.0
		91.4	91.1	20	171	711	72.0	97.7
SUSTech	R1	98.8	98.5	5	6	145	77.3	99.9
		98.8	98.5	10	12	137	87.8	99.9
		98.8	98.5	20	24	122	94.2	99.7
	R4	98.4	98.6	5	8	131	79.0	99.9
		98.4	98.6	10	16	124	88.8	99.8
		98.4	98.6	20	32	110	94.7	99.6

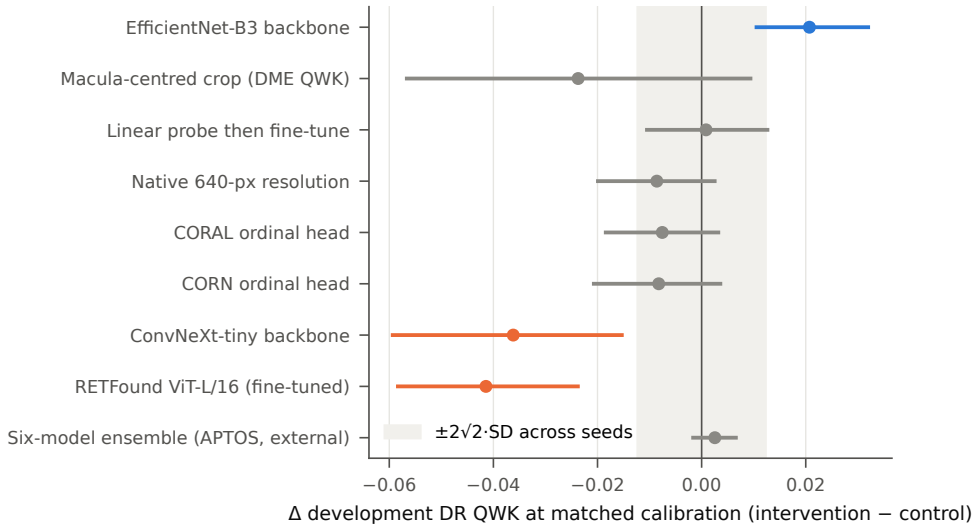


Figure S1: Pre-registered interventions on the model. Difference in development DR QWK (intervention minus control) under the matched decision rule with 95% paired-bootstrap intervals; the macula-centred crop was registered on the DME head and the ensemble on the external set. The grey band is $\pm 2\sqrt{2}$ standard deviations of matched development QWK across five seeds of the reference configuration (0.0123). Blue, improvement confirmed; orange, interval excludes zero in the unfavourable direction; grey, indistinguishable.

Supplementary Note 1. Decision rules

Every model emits, for DR, four logits whose sigmoids estimate $P(y > k)$, $k = 0, \ldots, 3$ (cumulative head and CORAL), or conditional probabilities $P(y > k \mid y > k-1)$ that are chained by cumulative product (CORN). From these the expected grade $s = \sum_k P(y > k)$ is computed.

R1 (shipped). Grade $= \#\{k : P(y > k) > 0.5\}$ after enforcing monotonicity by a running minimum.

R2 (development QWK cut-points). Four ordered cut-points on s maximising QWK on the development out-of-fold predictions, by coordinate ascent over 60 candidate values per cut-point (8 rounds), applied unchanged externally.

R3 (R2 with a referral constraint). R2 with the grade-1/2 cut-point replaced by the largest value that keeps development referable sensitivity at or above 0.90.

R4 and R4m (external, diagnostic). The same search on the external labels (80 candidates), with the external set split at random into two halves (seed 0); cut-points fitted on one half are applied to the other. R4 maximises QWK, R4m macro-averaged recall.

R5 (EM prior adaptation). Class probabilities $\pi_k(x)$ are derived from the monotone cumulative outputs ($\pi_0 = 1 - p_0$, $\pi_k = p_{k-1} - p_k$, $\pi_4 = p_3$), clipped and renormalised. The source prior is the mean development out-of-fold class probability; the target prior is estimated on the unlabelled external images by the EM procedure of Saerens *et al.*; posteriors are reweighted by the ratio of the priors and decoded by argmax.

R5b (black-box shift estimation). Importance weights $w = C^{-1}\mu$, with C the development confusion matrix of R1 and μ the external distribution of R1 predictions (negative weights set to zero); posteriors reweighted and decoded as for R5.

R4/R4m use external labels and are never reported as performance. Changing the development grid from 60 to 80 values changes the reference model's matched development QWK by 0.0033; 60 is used for all development quantities because it reproduces the archived matched comparisons exactly. Argmax decoding of the class probabilities derived from cumulative outputs never produced a mild prediction for R5 on APTOS or DDR, which limits what prior adaptation can do for the mild grade whatever the prior estimate.

Supplementary Note 2. Macular oedema: a supervision ceiling and an underivable grade

No intervention moved the three-grade DME head. A crop centred on the fovea by a localiser validated before use (median error 0.196 disc diameters), which encodes the clinical definition of the grade, changed DME QWK by −0.0237 [−0.0566, +0.0094]. Only 51 of the 516 three-grade images (9.9%) carry the middle grade, and a deployed commercial system reached a DME sensitivity of 26.5% (κ 0.38) on the same images on which its referable-DR κ was 0.81 (Duggal *et al.*, JMIR Med Inform 2025).

IDRiD defines its DME risk grade by the distance from the fovea to the nearest hard exudate: grade 0, no exudate; grade 1, exudates more than one disc diameter from the fovea; grade 2, within one disc diameter. Because the definition is geometric, a grade derived from the pixel-level exudate masks that IDRiD releases for 81 images, with its fovea and optic-disc annotations, should reproduce the expert grade if the grade is what its definition says; the test was committed before it was computed (Table S1). Of the 81 images, 54 could be linked to their grading-set copies (minimum pixel correlation 0.9999). The derived grade matched the expert grade in 96.3% of images, against 92.6% for a constant predictor, because almost every segmentation image is grade 2. On the 4 informative images (expert grade 0 or 1) the derivation failed twice, each time on a single small

exudate lying just inside the threshold, so the pre-specified gate failed and no derived DME labels were used.

Supplementary Note 3. A per-grade verdict that depends on the cut-point procedure

A development source control (E12) compared runs identical except for whether Messidor-2 images came from the default or the native-resolution files, on the 953 development images both could use. Moderate recall, native minus default, under four ways of setting both arms' cut-points:

Cut-points for both arms	Moderate recall difference (points) [95% CI]
Shipped (0.5)	−6.9 [−12.1, −2.1]
Fitted on all rows (in-sample)	−10.8 [−16.2, −5.4]
Random two-fold cross-fit, grid 80	−6.6 [−12.0, −1.1]
Fold-wise cross-fit, grid 60 (development standard)	+0.4 [−5.7, +6.1]

The originally archived value, cited in the project as a difference that *survived* matched cut-points, is reproduced by the random-half procedure; under the fold-wise procedure used for every other development comparison it dissolves. A matched-decision-rule claim must therefore name its fitting procedure, and per-grade claims should be reported under at least two.

Supplementary Note 4. Correction of an archived ordinal-head comparison

Regenerating every development number from the archived per-image logits with independent code reproduced each archived matched QWK exactly except one. The archived comparison of the CORN head against the reference reported −0.0479 [−0.0656, −0.0325], with matched QWK 0.8168 for CORN. The archived script had decoded CORN's conditional logits as if they were cumulative; decoding them by cumulative product gives 0.8564 and a difference of −0.0082 [−0.0207, +0.0036], indistinguishable from the reference. The pre-registered verdict (no improvement) is unchanged; the claim that CORN is worse is withdrawn.

Supplementary Note 5. Seed variation (O1–O3)

The reference configuration was trained with 5 seeds on identical data, split and recipe (O1, the noise floor): matched development QWK 0.8646–0.8764 (s.d. 0.0043); APTOS QWK at the shipped rule 0.8900–0.8968 (s.d. 0.0026). Of 50 per-grade recall differences between pairs of seeds on APTOS, 15 had an interval excluding zero at the shipped rule (largest 8.2 points) and 29 with refitted cut-points (O2 held); on DDR 30 and 35 of 50. Mild recall on APTOS was 5.4–6.2% at the shipped rule and 63.8–72.7% under R4m in every seed (O3 held).

Table S9: Seed variation of the reference configuration (O1–O3). Five independent training seeds of E08 (identical data, split and recipe).

Seed	Dev QWK (matched)	APTOS QWK (R1)	APTOS mild R1 (%)	APTOS mild R4m (%)	DDR QWK (R1)
E08	0.8646	0.897	5.9	70.8	0.733
S1E08	0.8685	0.896	5.4	70.3	0.717
S2E08	0.8714	0.895	6.2	72.7	0.704
S3E08	0.8764	0.890	5.9	72.4	0.728
S4E08	0.8692	0.894	5.9	63.8	0.713

Supplementary Note 6. Reverse direction (O4–O6)

The reference recipe was trained on APTOS 2019 (own frozen 5-fold split, near-duplicates grouped) and evaluated on Messidor-2 and IDRiD. Matched out-of-fold QWK on APTOS was 0.9134. On Messidor-2 at the shipped rule mild recall was 34.4% (O4 was falsified); external cut-points raised it to 44.4% (O5 was falsified). The mean predicted-minus-true grade at the shipped rule was −0.26 (O6, under-grading, held).

Table S10: Reverse direction (O4–O6). The reference recipe trained on APTOS 2019 and evaluated on Messidor-2 and IDRiD.

Corpus	Rule	QWK	Mild recall (%)	Mild F1	Sens. (%)	Spec. (%)	Pred.−true grade
Messidor-2	R1	0.715	34.4	0.276	43.5	99.2	−0.26
	R2	0.705	16.3	0.168	47.3	98.9	−0.32
	R3	0.679	17.0	0.155	34.4	99.7	−0.36
	R4	0.779	25.9	0.273	77.9	93.8	−0.02
	R4m	0.755	44.4	0.325	82.7	91.7	+0.16
IDRiD	R1	0.807	64.0	0.239	81.7	96.9	−0.24
	R2	0.827	36.0	0.200	85.4	95.3	−0.23
	R3	0.810	44.0	0.173	75.5	97.9	−0.31
	R4	0.841	28.0	0.200	88.2	92.7	−0.04
	R4m	0.828	56.0	0.231	88.9	92.2	+0.08

Supplementary Note 7. Content-overlap audit

Every image of the development pool, APTOS, DDR and EyePACS was reduced to a 256-bit difference hash after the training crop. In this project's data, re-encodings of one photograph lie within 8 bits of each other and distinct photographs at least 28 bits apart; all pairs within 16 bits were counted and every pair within 8 bits was archived.

Development pool and external sets. No development image lay within 8 bits of any APTOS or DDR image (closest pairs 12 and 14 bits).

EyePACS and external sets. No external image was hash-identical to an EyePACS image. 18 APTOS–EyePACS pairs (11 APTOS images) and 2,304 DDR–EyePACS pairs (202 DDR images) lay within 8 bits, the closest at 6 and 3 bits; their distances rise towards the threshold rather than peaking at zero, as expected of hash collisions between dark, low-contrast photographs. Removing every flagged image changed no model's shipped-rule referable sensitivity, mild recall, QWK or referral AUC by more than 0.55 percentage points on DDR and 0.04 on APTOS.

Within corpora. The development pool contains 10 hash-identical pairs, 7 of which straddle two folds of the frozen split (14 images, missed when the split was built); removing them changes every model's matched development QWK by at most 0.0008. APTOS and DDR contain 148 and 120 hash-identical pairs.

Table S11: Content-overlap audit. Pairs of images across corpora whose 256-bit difference hashes lie within the stated Hamming distance.

Corpus pair	Images	Pairs ≤16 bits	≤8 bits	0 bits
within Development pool	2,260	29	10	10
Development pool vs APTOS	2,260 / 3,662	47	0	0
within APTOS	3,662	1,355	149	148
Development pool vs DDR	2,260 / 12,522	14	0	0
within DDR	12,522	23,021	265	120
APTOS vs DDR	3,662 / 12,522	1,361	1	0
EyePACS vs APTOS	35,126 / 3,662	10,086	18	0
EyePACS vs DDR	35,126 / 12,522	267,402	2,304	0
EyePACS vs Development pool	35,126 / 2,260	178	0	0
Threshold 16 bits; development pairs split across folds: 7				

Full decision-rule matrices

Table S12: Full decision-rule matrix, APTOS. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.897	73.5	97.0	5.9	64.2	64.2	52.2	99.5	84.3
	R2	0.883	68.2	98.3	8.9	39.0	56.0	65.4	99.1	85.1
	R3	0.882	67.9	98.3	4.6	39.2	56.0	65.4	99.3	84.2
	R4	0.904	78.2	98.8	18.1	77.9	49.7	47.8	96.6	87.0
	R4m	0.886	75.2	96.6	70.8	49.0	47.2	56.3	80.4	95.4
E09	R1	0.879	69.3	88.5	4.3	66.8	62.7	45.8	99.3	84.1
	R2	0.863	62.0	88.5	22.7	30.3	62.2	56.6	96.3	87.4
	R3	0.847	61.0	88.5	1.4	34.3	62.2	56.6	99.7	79.7
	R4	0.898	76.7	98.6	10.0	75.5	48.7	48.5	98.6	85.9
	R4m	0.886	75.6	97.8	61.6	53.6	50.3	48.5	81.1	94.7
E10	R1	0.887	70.8	94.4	4.3	59.9	72.5	45.8	99.0	84.2
	R2	0.885	70.1	98.1	7.0	50.3	42.0	63.1	98.8	84.7
	R3	0.884	69.8	98.1	3.2	50.8	42.0	63.1	99.2	84.0
	R4	0.898	76.8	98.0	13.5	75.5	50.3	49.2	97.2	85.9
	R4m	0.876	73.0	97.8	77.8	36.6	50.8	52.5	72.6	97.0
E11FULL	R1	0.903	74.9	98.1	4.9	65.6	63.2	60.3	99.3	84.3
	R2	0.900	72.9	98.9	7.6	54.5	63.2	64.1	99.0	84.9
	R3	0.900	72.8	98.9	7.0	54.6	63.2	64.1	99.1	84.8
	R4	0.907	79.1	98.6	16.8	81.8	39.4	54.6	96.5	86.6
	R4m	0.890	73.4	97.6	75.1	37.3	51.3	59.7	75.3	96.4
E14MAC	R1	0.880	68.7	95.5	5.7	53.9	72.5	31.9	98.7	84.3
	R2	0.879	67.2	98.6	5.7	39.6	72.0	42.7	98.3	84.7
	R3	0.877	67.1	98.6	1.4	40.9	72.0	42.7	99.2	83.6
	R4	0.889	74.6	98.9	10.3	69.0	52.8	40.3	97.6	85.7
	R4m	0.857	71.5	97.4	75.7	35.7	44.0	46.8	71.2	96.6
E15LPFT	R1	0.874	65.3	89.6	5.1	45.1	70.5	57.6	99.2	84.2
	R2	0.880	69.2	97.2	10.3	43.5	57.0	66.1	98.3	85.3
	R3	0.878	68.9	97.2	5.1	44.5	57.0	66.1	99.1	84.2
	R4	0.898	77.3	98.6	11.9	77.5	44.6	49.2	97.3	86.0
	R4m	0.878	75.5	97.7	70.3	49.3	34.2	61.4	78.5	96.0
E17NAT	R1	0.891	71.1	96.1	5.9	56.8	72.0	47.8	99.2	84.6
	R2	0.872	65.5	97.3	9.5	32.2	48.2	64.7	99.0	85.0
	R3	0.870	64.9	97.3	3.2	32.6	48.2	64.7	99.4	83.7
	R4	0.894	76.5	99.1	16.2	72.0	48.7	47.8	96.3	86.9
	R4m	0.874	73.6	97.7	72.7	41.7	50.3	51.2	74.0	96.3
E20CORAL	R1	0.879	66.3	96.1	14.3	39.1	64.8	42.4	98.2	85.3
	R2	0.869	65.7	98.1	13.8	33.6	47.7	53.2	98.2	85.4
	R3	0.867	65.2	98.1	4.1	35.2	47.7	53.2	99.4	83.3
	R4	0.887	75.2	97.8	15.7	73.8	35.8	42.4	97.1	85.7
	R4m	0.872	73.1	97.3	72.4	45.4	48.2	35.6	79.4	95.5
E22CORN	R1	0.892	72.7	97.6	1.6	62.3	71.0	45.8	99.3	83.9
	R2	0.889	69.3	97.0	10.8	47.3	76.2	43.1	98.4	85.3
	R3	0.887	69.1	97.0	5.4	48.5	76.2	43.1	99.3	84.0
	R4	0.896	77.2	98.2	14.6	76.3	54.4	44.7	96.9	86.4
	R4m	0.875	75.0	96.8	77.0	48.9	43.0	47.8	74.1	96.7
E21CNXA	R1	0.862	62.9	84.7	8.9	48.5	69.4	41.7	98.7	84.5
	R2	0.860	62.0	96.0	13.5	21.4	62.7	52.2	97.5	85.8
	R3	0.855	61.3	96.0	1.4	23.4	62.7	52.2	99.2	82.3
	R4	0.890	76.1	97.8	18.6	74.4	51.3	37.3	96.8	87.1
	R4m	0.873	74.3	98.0	67.0	48.3	51.8	41.0	79.6	95.3

Table S13: Full decision-rule matrix, DDR. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.733	73.7	97.7	33.3	51.2	58.5	50.3	65.5	98.7
	R2	0.734	68.9	98.8	22.1	38.2	53.4	50.3	62.5	99.0
	R3	0.742	70.5	98.8	17.5	43.4	53.4	50.3	67.1	98.3
	R4	0.805	69.8	86.2	25.2	57.8	57.2	49.2	80.8	93.2
	R4m	0.787	67.0	83.6	59.8	49.8	55.9	45.3	66.8	98.4
E09	R1	0.697	68.8	93.5	20.5	46.4	54.7	46.3	63.9	97.3
	R2	0.708	58.6	92.0	31.7	18.1	56.8	47.0	50.4	99.2
	R3	0.733	66.5	92.0	11.7	43.0	56.8	47.0	72.1	93.4
	R4	0.753	60.7	77.1	38.4	43.6	48.3	50.4	70.9	94.1
	R4m	0.744	57.3	71.6	51.6	41.2	61.0	41.4	64.9	96.9
E10	R1	0.737	70.5	96.1	43.7	44.8	59.3	42.7	62.8	98.2
	R2	0.740	69.3	97.4	27.1	41.9	41.9	47.8	63.4	98.2
	R3	0.748	71.1	97.4	21.6	47.6	41.9	47.8	68.3	97.4
	R4	0.793	67.2	85.1	23.5	53.1	67.4	43.3	80.4	92.6
	R4m	0.775	63.4	83.0	57.9	40.9	58.5	43.4	66.7	97.6
E11FULL	R1	0.755	76.1	98.2	29.8	56.7	55.5	56.1	70.3	98.7
	R2	0.756	72.5	98.9	22.1	46.9	56.4	55.6	65.2	99.0
	R3	0.757	72.6	98.9	21.4	47.4	56.4	55.6	65.6	98.9
	R4	0.813	72.8	89.2	24.6	60.2	60.6	58.4	81.2	94.2
	R4m	0.799	66.7	84.2	58.7	46.3	64.8	53.1	69.4	98.4
E14MAC	R1	0.736	70.2	96.3	42.1	43.7	71.2	39.6	61.7	98.9
	R2	0.729	68.1	98.6	19.5	37.1	62.7	45.2	62.5	98.8
	R3	0.740	70.4	98.6	10.2	44.9	62.7	45.2	69.4	97.4
	R4	0.793	67.8	84.7	24.0	55.2	61.9	45.5	80.1	93.1
	R4m	0.776	64.8	84.4	57.1	43.8	68.2	38.4	65.6	98.3
E15LPFT	R1	0.764	69.9	92.5	45.2	43.7	63.1	62.3	66.1	98.3
	R2	0.743	68.9	97.5	21.4	38.2	51.7	60.0	62.2	98.8
	R3	0.753	71.2	97.5	16.5	45.4	51.7	60.0	68.4	97.9
	R4	0.794	67.5	84.6	35.2	52.0	66.5	48.6	76.8	94.5
	R4m	0.786	64.4	81.0	57.6	44.5	61.0	54.0	67.6	97.9
E17NAT	R1	0.743	71.5	96.9	38.9	46.4	58.9	45.5	64.6	98.2
	R2	0.765	66.9	96.1	33.8	35.0	54.7	48.5	65.2	98.3
	R3	0.772	68.9	96.1	26.5	41.6	54.7	48.5	71.0	97.0
	R4	0.801	67.5	85.7	31.4	51.8	55.5	48.2	79.4	93.4
	R4m	0.789	63.4	83.3	62.1	40.4	63.6	41.1	67.5	97.8
E20CORAL	R1	0.740	65.2	93.8	37.8	36.0	59.3	33.1	58.9	98.5
	R2	0.728	65.6	96.2	24.9	33.8	50.8	43.8	58.9	98.5
	R3	0.743	69.6	96.2	14.4	46.5	50.8	43.8	69.7	96.6
	R4	0.778	64.2	82.9	37.1	46.4	57.6	43.5	75.1	94.1
	R4m	0.768	60.6	81.0	58.7	37.1	64.4	35.5	65.6	97.7
E22CORN	R1	0.699	72.4	98.7	25.9	49.1	56.4	42.5	63.8	98.6
	R2	0.707	66.5	98.0	32.4	34.7	64.0	30.0	54.7	99.3
	R3	0.730	70.2	98.0	24.0	46.3	64.0	30.0	65.0	98.3
	R4	0.785	67.4	85.2	22.7	54.5	56.4	41.9	79.4	92.8
	R4m	0.766	63.6	85.9	56.5	39.1	63.1	36.3	63.2	98.6
E21CNXA	R1	0.710	65.4	88.0	34.9	45.0	51.7	34.7	62.6	98.4
	R2	0.701	62.1	94.9	23.8	26.3	53.8	40.4	55.1	99.2
	R3	0.726	67.8	94.9	8.7	44.5	53.8	40.4	71.3	95.8
	R4	0.747	58.7	78.5	41.4	37.3	56.4	40.3	70.0	96.3
	R4m	0.738	56.9	70.5	53.5	43.0	59.3	32.5	67.4	97.3

Table S14: Full decision-rule matrix, DeepDRiD. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.792	64.3	90.8	19.8	75.4	19.8	37.5	86.6	93.9
	R2	0.812	64.0	93.5	10.8	58.5	35.9	37.5	83.7	94.7
	R3	0.815	64.5	93.5	6.3	63.3	35.9	37.5	86.6	93.4
	R4	0.833	60.1	80.6	33.8	49.5	44.1	33.0	86.9	93.0
	R4m	0.818	61.5	73.0	40.5	64.1	58.2	10.7	87.0	92.4
E09	R1	0.747	60.8	92.3	10.8	57.3	22.9	34.8	77.8	95.1
	R2	0.757	57.2	89.6	22.5	23.9	39.5	35.7	63.1	97.9
	R3	0.777	61.6	89.6	7.2	54.3	39.5	35.7	83.1	90.4
	R4	0.793	57.9	81.2	31.5	28.1	55.9	32.1	76.5	95.2
	R4m	0.781	57.0	75.3	32.0	46.7	45.5	29.5	77.1	93.1
E10	R1	0.797	63.4	87.9	32.0	66.8	24.6	37.5	83.8	93.9
	R2	0.808	63.4	90.6	18.5	62.8	29.9	39.3	83.9	93.9
	R3	0.813	63.7	90.6	12.6	67.6	29.9	39.3	87.7	92.0
	R4	0.820	59.7	78.7	35.6	54.3	38.1	40.2	86.7	92.3
	R4m	0.806	56.5	63.2	55.4	54.8	51.1	27.7	83.8	93.9
E11FULL	R1	0.819	66.5	90.7	23.0	81.7	19.8	48.2	90.3	92.3
	R2	0.828	65.5	91.6	18.9	71.9	26.3	45.5	87.6	93.1
	R3	0.830	65.5	91.6	18.9	72.4	26.3	45.5	88.1	93.1
	R4	0.844	66.3	83.2	31.5	65.6	57.9	26.8	91.4	91.1
	R4m	0.839	61.8	79.2	41.0	65.3	30.8	45.5	87.8	93.0
E14MAC	R1	0.777	64.3	89.6	28.8	67.1	31.1	24.1	81.1	95.6
	R2	0.788	64.8	94.3	11.3	58.5	41.8	24.1	81.7	95.9
	R3	0.791	65.3	94.3	5.4	64.6	41.8	24.1	86.6	92.7
	R4	0.805	58.6	80.7	34.2	39.7	48.0	26.8	83.0	94.2
	R4m	0.797	59.9	75.9	40.1	59.5	42.9	23.2	84.3	94.0
E15LPFT	R1	0.794	63.9	88.8	24.8	62.8	30.5	48.2	85.1	92.9
	R2	0.799	63.0	92.3	11.7	61.3	27.7	43.8	83.2	93.8
	R3	0.801	63.6	92.3	8.6	65.8	27.7	43.8	86.1	92.3
	R4	0.822	62.8	84.0	26.1	54.0	48.9	37.5	87.5	90.7
	R4m	0.809	57.8	73.0	43.7	57.5	31.9	44.6	84.6	92.8
E17NAT	R1	0.797	63.2	86.0	34.2	67.6	26.3	36.6	84.7	93.5
	R2	0.826	63.1	85.6	29.3	55.0	43.2	39.3	86.1	93.0
	R3	0.823	63.0	85.6	18.9	60.1	43.2	39.3	89.1	89.6
	R4	0.822	59.5	79.5	36.9	41.5	48.3	39.3	83.1	94.0
	R4m	0.816	57.4	71.8	42.8	53.0	39.0	42.0	85.4	93.3
E20CORAL	R1	0.798	63.2	89.6	25.2	52.8	44.1	20.5	78.7	95.6
	R2	0.797	63.5	92.6	18.9	49.2	43.2	29.5	78.7	95.6
	R3	0.803	65.4	92.6	9.9	63.8	43.2	29.5	86.9	92.3
	R4	0.823	60.4	82.8	28.8	35.2	59.3	32.1	83.7	92.9
	R4m	0.795	53.7	55.8	60.4	50.5	55.9	27.7	84.6	93.5
E22CORN	R1	0.770	64.8	91.9	14.9	73.1	26.0	35.7	85.2	92.6
	R2	0.794	65.0	90.2	23.9	58.3	45.5	27.7	78.9	95.7
	R3	0.798	65.8	90.2	14.9	67.1	45.5	27.7	85.1	91.9
	R4	0.818	62.4	83.0	37.4	46.5	51.1	34.8	82.3	94.2
	R4m	0.813	62.2	75.8	42.3	59.5	52.8	29.5	83.0	93.8
E21CNXA	R1	0.778	61.9	89.3	24.8	60.1	27.1	27.7	80.8	94.6
	R2	0.790	61.6	92.6	18.5	38.7	43.2	33.9	75.8	96.4
	R3	0.798	63.6	92.6	7.7	55.0	43.2	33.9	86.5	91.5
	R4	0.811	58.5	80.7	27.5	39.2	51.7	27.7	86.6	91.0
	R4m	0.796	60.0	72.2	44.1	56.0	58.5	10.7	82.9	93.6

Table S15: Full decision-rule matrix, SUSTech. QWK, accuracy, per-grade recall (%) and referable-DR sensitivity/specificity (%) of every model under every rule. R4/R4m are fitted on the external labels by two-fold cross-fitting and are diagnostic bounds, not performance.

Model	Rule	QWK	Acc.	No DR	Mild	Mod.	Sev.	PDR	Sens.	Spec.
E08	R1	0.923	82.0	96.4	33.3	64.7	94.5	39.7	99.0	97.9
	R2	0.904	74.2	97.5	25.0	43.0	53.4	63.8	98.2	98.3
	R3	0.902	74.4	97.5	12.5	44.4	53.4	63.8	99.2	97.3
	R4	0.932	84.4	98.1	25.0	73.4	90.4	22.4	97.4	98.6
	R4m	0.900	76.7	82.9	79.2	70.4	82.2	41.4	92.5	99.5
E09	R1	0.892	78.6	90.6	29.2	64.9	87.7	43.1	99.2	93.9
	R2	0.890	69.5	88.7	58.3	37.8	74.0	58.6	97.4	98.6
	R3	0.857	69.5	88.7	12.5	40.8	74.0	58.6	99.6	89.3
	R4	0.925	85.2	96.2	20.8	80.5	87.7	19.0	97.4	98.5
	R4m	0.912	81.2	89.5	50.0	75.3	71.2	53.4	97.4	98.3
E10	R1	0.929	80.5	97.8	50.0	55.3	97.3	41.4	98.6	98.8
	R2	0.901	76.2	99.5	33.3	50.1	26.0	67.2	98.2	98.8
	R3	0.903	76.5	99.5	33.3	51.0	26.0	67.2	99.0	98.8
	R4	0.937	87.5	99.4	25.0	84.9	71.2	20.7	99.2	98.5
	R4m	0.918	81.5	95.7	75.0	62.2	84.9	46.6	90.1	99.7
E11FULL	R1	0.929	83.2	98.1	33.3	64.9	94.5	43.1	98.8	98.5
	R2	0.924	79.7	99.2	20.8	54.2	79.5	51.7	98.4	98.9
	R3	0.924	79.7	99.2	16.7	54.5	79.5	51.7	98.6	98.8
	R4	0.934	87.3	98.4	20.8	83.3	80.8	27.6	98.4	98.6
	R4m	0.919	81.4	92.9	62.5	69.9	78.1	41.4	95.8	99.8
E14MAC	R1	0.919	77.3	95.7	58.3	51.5	95.9	24.1	98.2	99.2
	R2	0.916	75.6	99.7	20.8	40.5	86.3	43.1	98.2	99.1
	R3	0.917	75.5	99.7	4.2	41.4	86.3	43.1	99.0	98.3
	R4	0.928	84.3	99.2	20.8	74.0	78.1	20.7	97.6	99.1
	R4m	0.916	80.0	89.9	75.0	71.5	74.0	36.2	96.6	99.7
E15LPFT	R1	0.910	75.2	93.2	41.7	47.9	82.2	56.9	98.4	98.2
	R2	0.906	75.8	98.3	20.8	44.7	60.3	70.7	97.4	98.9
	R3	0.902	76.1	98.3	8.3	46.3	60.3	70.7	98.6	97.4
	R4	0.936	85.2	98.1	20.8	77.0	74.0	37.9	98.0	98.2
	R4m	0.920	79.6	90.6	75.0	65.2	75.3	56.9	95.8	99.7
E17NAT	R1	0.929	80.1	97.9	37.5	54.2	98.6	43.1	99.0	98.8
	R2	0.895	70.2	98.7	29.2	32.9	26.0	67.2	98.0	98.8
	R3	0.897	70.7	98.7	20.8	35.1	26.0	67.2	99.8	98.3
	R4	0.937	87.1	99.4	12.5	81.1	83.6	27.6	99.8	98.3
	R4m	0.917	80.4	91.9	58.3	66.6	76.7	55.2	95.4	99.2
E20CORAL	R1	0.909	71.9	93.3	50.0	41.1	71.2	43.1	97.6	98.9
	R2	0.902	72.1	97.5	41.7	38.4	37.0	65.5	97.6	98.9
	R3	0.899	72.4	97.5	16.7	40.8	37.0	65.5	99.4	97.1
	R4	0.934	85.3	98.3	8.3	81.6	72.6	15.5	98.8	98.3
	R4m	0.887	70.9	73.9	70.8	74.0	58.9	34.5	95.8	99.5
E22CORN	R1	0.920	82.9	97.6	16.7	66.3	91.8	43.1	99.2	97.1
	R2	0.919	77.4	96.2	29.2	50.7	91.8	43.1	97.6	98.5
	R3	0.919	77.7	96.2	12.5	52.6	91.8	43.1	99.2	97.3
	R4	0.929	85.6	97.5	8.3	81.4	86.3	13.8	98.0	98.3
	R4m	0.906	78.0	84.8	70.8	72.6	71.2	50.0	95.2	99.5
E21CNXA	R1	0.883	65.3	77.7	41.7	46.0	80.8	43.1	98.8	98.0
	R2	0.889	65.6	91.9	37.5	26.0	49.3	60.3	98.0	99.1
	R3	0.876	65.9	91.9	12.5	28.8	49.3	60.3	100.0	94.7
	R4	0.926	83.3	98.4	4.2	76.2	76.7	5.2	98.4	98.8
	R4m	0.900	74.2	88.1	62.5	58.1	60.3	46.6	94.4	99.7